Multiple Regression Analysis: Further Issues

Intermediate Econometrics

Chengye Jia

Department of Economics
Shandong University of Finance and Economics

Spring 2026

Wooldridge, Chapter 6

Outline

Data Scaling and Beta Coefficients (§6-1)

Data Scaling and Beta Coefficients: Motivation & Setup

Motivation. Does it matter whether wage is in dollars or thousands, or birth weight in ounces or pounds? Rescaling a variable changes the *numbers* OLS reports — coefficients, standard errors, t, F, confidence intervals — yet should change **nothing essential**. Often we rescale purely for cosmetics (fewer leading zeros).

Setup. Track how each statistic transforms when y or x_j is multiplied by a constant, and confirm that measured effects and test outcomes are preserved. Then rescale to **standard-deviation units** — *beta (standardized) coefficients* — to compare regressors on a common footing. **Plan:** rescaling y, rescaling x_j, log invariance, beta coefficients.

Effects of Rescaling the Dependent Variable

Will switching birth weight from *ounces* to *pounds* change any conclusion? Rescale *y* and watch every statistic. Consider the birth weight equation

$$\widehat{bwght} = \underset{(1.049)}{116.974} - \underset{(0.0916)}{0.4634}\, cigs + \underset{(0.0292)}{0.0927}\, faminc,$$

and define $bwghtlbs = bwght/16$ (ounces $\to$ pounds).

What changes? Every coefficient and standard error is divided by 16, so the t statistics and R^2 are **unchanged**; SSR falls by $16^2 = 256$ and the SER by 16.

Rule — rescaling *y* by *c*: $\hat{\beta}_j \to c\,\hat{\beta}_j$ and $\mathrm{se}(\hat{\beta}_j) \to c\ \mathrm{se}(\hat{\beta}_j)$ (so $t_{\hat{\beta}_j}$ and R^2 are unchanged); $\text{SSR} \to c^2\text{SSR}$, $\text{SER} \to |c|\,\text{SER}$.

Key insight: rescaling *y* is purely cosmetic — it changes the appearance, not the conclusions.

Effects of Rescaling an Independent Variable

Now define $packs = cigs/20$ (cigarettes $\to$ packs per day). Rewriting the equation,

$$\widehat{bwght} = \underset{(1.049)}{116.974} - \underset{(1.832)}{9.268}\, packs + \underset{(0.0292)}{0.0927}\, faminc.$$

What changes? Only the *packs* terms: its coefficient becomes $20 \times (-0.4634) = -9.268$ and its standard error $20 \times 0.0916 = 1.832$, leaving the t statistic $-9.268/1.832 = -5.06$ **unchanged**. The intercept, the other coefficients, R^2, SSR, and SER are all unchanged.

> **Rule — rescaling x_j by c:** $\hat{\beta}_j \to \hat{\beta}_j/c$ and $\mathrm{se}(\hat{\beta}_j) \to \mathrm{se}(\hat{\beta}_j)/c$ (so $t_{\hat{\beta}_j}$ is unchanged); the other coefficients, R^2, SSR, and SER are all unchanged.

Bottom line. Rescaling any variable changes coefficients and standard errors proportionally, but never the t statistics, F statistics, R^2, or any test conclusion.

Rescaling Summary and Logarithmic Invariance

	Rescale y ($y \to cy$)	**Rescale** x_j ($x_j \to cx_j$)
$\hat{\beta}_j$	$\times c$	$\div c$
$\mathrm{se}(\hat{\beta}_j)$	$\times c$	$\div c$
$t_{\hat{\beta}_j}$	unchanged	unchanged
R^2	unchanged	unchanged
SSR	$\times c^2$	unchanged
SER	$\times \lvert c \rvert$	unchanged

Logarithmic invariance:

If y appears as $\log(y)$: changing the units of y only affects the intercept.

$\log(c \cdot y) = \log(c) + \log(y)$

All slope coefficients are **unchanged**.

Similarly, if x_j appears as $\log(x_j)$: changing the units of x_j only shifts the intercept.

Elasticities are invariant to units of measurement.

Practical advice. Use rescaling to make coefficients easy to read — eliminate long strings of zeros (if $\hat{\beta} = 0.00013$, measure x in thousands). This is cosmetic, not substantive.

Beta (Standardized) Coefficients: Definition

When regressors live on wildly different scales — dollars, years, test points — the raw $\hat{\beta}_j$ cannot be ranked by "importance": a small coefficient may merely reflect large units. **Standardizing** puts every variable in common units.

Beta Coefficients

Standardize all variables to *z*-scores:

$$z_y = \frac{y_i - \bar{y}}{\hat{\sigma}_y}, \qquad z_j = \frac{x_{ij} - \bar{x}_j}{\hat{\sigma}_j}$$

Then run the regression (no intercept needed):

$$\hat{z}_y = \hat{b}_1 z_1 + \hat{b}_2 z_2 + \cdots + \hat{b}_k z_k$$

The **beta coefficient** (standardized coefficient) is:

$$\hat{b}_j = \frac{\hat{\sigma}_j}{\hat{\sigma}_y}\hat{\beta}_j \quad \text{for } j = 1, \ldots, k$$

Where this comes from. Write the fitted equation in deviations, $\hat{y}_i - \bar{y} = \sum_j \hat{\beta}_j (x_{ij} - \bar{x}_j)$; divide by $\hat{\sigma}_y$ and multiply/divide term j by $\hat{\sigma}_j$:

$$\frac{\hat{y}_i - \bar{y}}{\hat{\sigma}_y} = \sum_j \underbrace{\frac{\hat{\sigma}_j}{\hat{\sigma}_y}\hat{\beta}_j}_{\hat{b}_j} \frac{x_{ij} - \bar{x}_j}{\hat{\sigma}_j}.$$

So $\hat{b}_j$ is simply $\hat{\beta}_j$ rescaled by the ratio of standard deviations.

Beta Coefficients: Interpretation

Interpretation: A one *standard deviation* increase in x_j changes $\hat{y}$ by $\hat{b}_j$ standard deviations.

This puts all variables on the same scale, allowing direct comparison of the relative importance of different regressors.

When are beta coefficients useful?

- Variables measured on different scales (test scores, income, age)
- Want to rank variables by "importance"
- Especially useful when units are arbitrary or hard to interpret

Key facts:

- The t statistics are **identical** for standardized and unstandardized variables
- With one regressor: the beta coefficient equals the sample correlation r_{xy} (between -1 and 1)
- Even when elasticities are of interest, beta coefficients can still be informative

Example 6.1 — Effects of Pollution on Housing Prices

Beta Coefficients for Housing Prices (HPRICE2)

The level-level model:

$$\textit{price} = \beta_0 + \beta_1 \textit{nox} + \beta_2 \textit{crime} + \beta_3 \textit{rooms} + \beta_4 \textit{dist} + \beta_5 \textit{stratio} + u$$

Beta coefficient estimates (each variable replaced by its z-score):

$$\widehat{\textit{zprice}} = -0.340\,\textit{znox} - 0.143\,\textit{zcrime} + 0.514\,\textit{zrooms} - 0.235\,\textit{zdist} - 0.270\,\textit{zstratio}$$

Ranking by importance:

1. *rooms*: $|\hat{b}| = 0.514$ (largest)
2. *nox*: $|\hat{b}| = 0.340$
3. *stratio*: $|\hat{b}| = 0.270$
4. *dist*: $|\hat{b}| = 0.235$
5. *crime*: $|\hat{b}| = 0.143$ (smallest)

A one-sd increase in *nox* decreases price by 0.34 sd.

The same relative movement in pollution has a larger effect on price than crime.

Cannot conclude this from raw coefficients because units differ!

Logarithmic Functional Forms (§6-2a)

Logarithmic Functional Forms: Motivation & Setup

Motivation. Logs are everywhere in applied work: they deliver **constant-elasticity** and **semi-elasticity** interpretations that are *unit-free*, they narrow the range of variables, and they dampen the influence of outliers and heteroskedasticity — but they make no sense for variables that can be zero, negative, or are already proportions.

Setup. Compare the four forms (level–level, level–log, log–level, log–log) and their slope interpretations, then sharpen the *exact* percentage change $100[\exp(\hat{\beta}_1) - 1]$ against the approximation $100\hat{\beta}_1$. **Plan:** the four forms, a housing-price example, exact vs. approximate percentages, and rules of thumb for when to use logs.

Review: Four Functional Forms

The four functional form combinations for *y* and *x*:

Model	Dependent	Independent	Interpretation of β_1
Level–Level	y	x	$\Delta y = \beta_1 \Delta x$
Level–Log	y	$\log(x)$	$\Delta y = (\beta_1/100)\,\%\Delta x$
Log–Level	$\log(y)$	x	$\%\Delta y \approx 100\beta_1 \Delta x$
Log–Log	$\log(y)$	$\log(x)$	$\%\Delta y \approx \beta_1\,\%\Delta x$ (elasticity)

Remember: The log-level and log-log interpretations are *approximations*. For large changes, use the exact formula $\%\Delta y = 100[\exp(\hat{\beta}\Delta x) - 1]$ (next slide).

Functional Forms in Practice: Housing Prices

Housing Prices (HPRICE2 — eq. 6.7)

$$\widehat{\log(price)} = \underset{(0.19)}{9.23} - \underset{(0.066)}{0.718} \log(nox) + \underset{(0.019)}{0.306}\, rooms$$

$$n = 506, \quad R^2 = 0.514$$

Interpreting the coefficients:

- $\hat{\beta}_{\log(nox)} = -0.718$: elasticity
  1% increase in *nox* $\Rightarrow$ price falls by $\approx 0.718\%$
- $\hat{\beta}_{rooms} = 0.306$: semi-elasticity
  One more room $\Rightarrow$ price rises by $\approx 30.6\%$

> This equation mixes log–log (for *nox*) and log–level (for *rooms*).
>
> Both interpretations follow directly from the functional form table.
>
> The 30.6% approximation for *rooms* is somewhat rough — the exact change is 35.8% (next slide).

Exact vs. Approximate Percentage Changes

Exact Percentage Change (eq. 6.8)

In the model $\widehat{\log(y)} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2$, fixing x_1:

$$\%\Delta\hat{y} = 100 \cdot [\exp(\hat{\beta}_2 \Delta x_2) - 1]$$

When does this matter?

- For small $\hat{\beta}_2 \Delta x_2$: the approximation $100 \cdot \hat{\beta}_2 \Delta x_2$ is fine
- For large changes: the exact formula can differ substantially

Why? A first-order Taylor expansion $\exp(a) \approx 1 + a$ gives $100\,\hat{\beta}_2 \Delta x_2$, so it keeps only the *first-order* term; the dropped $a^2/2$ is why the exact increase and decrease are asymmetric.

Housing Prices: *rooms*

$\hat{\beta}_{rooms} = 0.306$

Add one room ($\Delta rooms = 1$):
Approx: 30.6%
Exact: $100[\exp(0.306) - 1] = 35.8\%$

Remove one room ($\Delta rooms = -1$):
Approx: -30.6%
Exact: $100[\exp(-0.306) - 1] = -26.4\%$

The approximation always lies between the exact increase and exact decrease.

When to Use Logarithms: Rules of Thumb

Use $\log(x)$ when:

- $x > 0$ always (wages, prices, sales, population)
- Large variation in x (firm sales span millions to billions)
- Want percentage-change interpretation
- Want to reduce the influence of outliers
- Want to improve normality of errors

Use x in levels when:

- x is measured in years (education, experience, age)
- x is a proportion or percent (unemployment rate, pass rate)
- x takes on zero or negative values

Benefits of logs:

1. Coefficients have percentage-change interpretation
2. Slope coefficients invariant to units
3. Narrows the range of variables $\Rightarrow$ less sensitivity to outliers
4. $\log(y)$ as dependent variable often makes CLM assumptions more plausible (homoskedasticity, normality)

Warning. If $y \geq 0$ with $y = 0$ possible, $\log(y)$ is undefined. The fix $\log(1 + y)$ is sometimes used, but its coefficients are **not** invariant to the units of y — use with caution.

Quadratic Models and Interactions (§6-2b, §6-2c)

Quadratics and Interactions: Motivation & Setup

Motivation. A linear model forces a *constant* partial effect, but economics often demands otherwise: the return to experience **diminishes**, and the payoff to class attendance may **depend on** ability. Quadratics and interactions relax the constant-effect straitjacket while staying linear in the parameters.

Setup. The partial effect is now itself a *function*, found by calculus: a quadratic gives $\partial y/\partial x = \beta_1 + 2\beta_2 x$, and an interaction gives $\partial y/\partial x_1 = \beta_1 + \beta_3 x_2$. **Plan:** quadratics and turning points, interaction terms, and worked examples.

Models with Quadratics

The simplest way to let a marginal effect *change* with the level of x — diminishing returns to experience, a U-shaped cost curve — is to add the square of x as its own regressor.

Quadratic Specification

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + u$$

The estimated equation: $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x + \hat{\beta}_2 x^2$.

The **marginal effect** of x on $\hat{y}$ depends on x:

$$\frac{\Delta \hat{y}}{\Delta x} \approx \hat{\beta}_1 + 2\hat{\beta}_2 x$$

Turning point. Set the marginal effect to zero, $\hat{\beta}_1 + 2\hat{\beta}_2 x^* = 0$:

$$x^* = -\frac{\hat{\beta}_1}{2\hat{\beta}_2} = \frac{|\hat{\beta}_1|}{2|\hat{\beta}_2|}$$

Check whether x^* is within the range of the data!

Two shapes:

- $\hat{\beta}_1 > 0$, $\hat{\beta}_2 < 0$: **hump shape** (diminishing returns)
- $\hat{\beta}_1 < 0$, $\hat{\beta}_2 > 0$: **U-shape** (increasing returns)
- Same sign on $\hat{\beta}_1$, $\hat{\beta}_2$: monotonic, no turning point for $x > 0$

Wage–Experience Quadratic (WAGE1)

Diminishing Returns to Experience (eq. 6.12)

$$\widehat{wage} = \underset{(0.35)}{3.73} + \underset{(0.041)}{0.298}\, exper - \underset{(0.0009)}{0.0061}\, exper^2$$

$$n = 526, \quad R^2 = 0.093$$

Marginal effect at different experience levels:

$$\frac{\Delta \widehat{wage}}{\Delta exper} \approx 0.298 - 2(0.0061)\, exper$$

- $exper = 1$: $0.298 - 0.012 = 0.286$ (\$0.29/hr)
- $exper = 10$: $0.298 - 0.122 = 0.176$ (\$0.18/hr)
- $exper = 20$: $0.298 - 0.244 = 0.054$ (\$0.05/hr)

Turning point:

$$exper^* = \frac{0.298}{2(0.0061)} \approx 24.4 \text{ years}$$

After 24.4 years, the estimated return to experience is **negative**.

About 28% of the sample has $exper > 24$, so the turning point is economically relevant — not just a mathematical artifact.

Example 6.2 — Housing Prices with Quadratic *rooms*

Quadratic in *rooms* (HPRICE2 — eq. 6.14)

$$\widehat{\log(price)} = \underset{(0.57)}{13.39} - \underset{(0.115)}{0.902}\log(nox) - \underset{(0.043)}{0.087}\log(dist)$$

$$-\underset{(0.165)}{0.545}\, rooms + \underset{(0.013)}{0.062}\, rooms^2 - \underset{(0.006)}{0.048}\, stratio$$

$$n = 506, \quad R^2 = 0.603$$

The U-shape: $\hat{\beta}_1 < 0$, $\hat{\beta}_2 > 0$. Turning point: $rooms^* = 0.545/[2(0.062)] \approx 4.4$. Only 1% of communities have ≤ 4.4 average rooms, so for the vast majority more rooms $\Rightarrow$ higher price.

Example 6.2 — Marginal Effect of *rooms*

Marginal effect of *rooms* on $\log(price)$:

$$\%\Delta\widehat{price} \approx (-54.5 + 12.4\,rooms)\,\Delta rooms$$

Effect at different values:

- At $rooms = 5$: $(-54.5 + 62.0) = 7.5\%$ per room
- At $rooms = 6$: $(-54.5 + 74.4) = 19.9\%$ per room
- At $rooms = 7$: $(-54.5 + 86.8) = 32.3\%$ per room

A very strong **increasing** effect at typical values!

Lesson: The coefficient on $rooms^2$ is just 0.062 — seemingly tiny. But the marginal effect $\hat{\beta}_1 + 2\hat{\beta}_2 \cdot rooms$ changes rapidly across the data range.

Never dismiss a quadratic term based on the size of the coefficient alone. Always compute the marginal effect at relevant values of *x*.

Important Lesson: Small Coefficients on x^2 Can Matter

Do not dismiss a quadratic term just because $\hat{\beta}_2$ "looks small."

In the housing example, $\hat{\beta}_{rooms^2} = 0.062$.
This seems tiny, but the marginal effect is:

$$\hat{\beta}_1 + 2\hat{\beta}_2 \cdot rooms$$

At *rooms* $= 6.45$: the slope is 0.255 (25.5% per room).
At *rooms* $= 7$: the slope is 0.323 (32.3% per room).

The quadratic term is highly significant ($t = 0.062/0.013 = 4.77$).

Compare with a linear model:

Dropping *rooms*^{2}, the coefficient on *rooms* becomes about 0.255, implying 25.5% per room regardless of current size.

The quadratic model reveals that the effect is **much stronger** for larger houses — an important insight the linear model misses entirely.

General rule: Always compute the marginal effect at interesting values of x (mean, quartiles).

Interaction Terms: When One Effect Depends on Another

Sometimes the effect of x_2 is not constant but **depends on the level of** x_1 — an extra bedroom is worth more in a larger house, and class attendance helps stronger students more. An *interaction term* $x_1 x_2$ captures this.

Interaction model and its partial effect

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3\, x_1 x_2 + u \quad \Longrightarrow \quad \frac{\partial y}{\partial x_2} = \beta_2 + \beta_3 x_1 .$$

The effect of x_2 now moves with x_1 (and symmetrically, $\partial y / \partial x_1 = \beta_1 + \beta_3 x_2$).

Reading β_2 correctly. β_2 is the effect of x_2 *only when* $x_1 = 0$ — often uninteresting or impossible (a house with zero square feet). Report the partial effect at **meaningful** values of x_1 — the mean, median, or quartiles — instead.

Example 6.3 — Attendance and Exam Performance: Model

Interactions with Quadratics (ATTEND — eq. 6.18–6.19)

$$stndfnl = \beta_0 + \beta_1\, atndrte + \beta_2\, priGPA + \beta_3\, ACT + \beta_4\, priGPA^2 + \beta_5\, ACT^2 + \beta_6\, priGPA \cdot atndrte +$$

$$\widehat{stndfnl} = \underset{(1.36)}{2.05} - \underset{(0.0102)}{0.0067}\, atndrte - \underset{(0.48)}{1.63}\, priGPA - \underset{(0.098)}{0.128}\, ACT$$

$$+\underset{(0.101)}{0.296}\, priGPA^2 + \underset{(0.0022)}{0.0045}\, ACT^2 + \underset{(0.0043)}{0.0056}\, priGPA \cdot atndrte$$

$$n = 680, \quad R^2 = 0.229, \quad \bar{R}^2 = 0.222$$

Example 6.3 — Partial Effect of Attendance

Partial effect of attendance on standardized final exam score:

$$\frac{\Delta \mathit{stndfnl}}{\Delta \mathit{atndrte}} = \hat{\beta}_1 + \hat{\beta}_6 \mathit{priGPA} = -0.0067 + 0.0056 \cdot \mathit{priGPA}$$

At different prior GPA levels:

- $\mathit{priGPA} = 2.0$: $-0.0067 + 0.0112 = 0.0045$
- $\mathit{priGPA} = 2.59$ (mean): $-0.0067 + 0.0145 = 0.0078$
- $\mathit{priGPA} = 3.5$: $-0.0067 + 0.0196 = 0.0129$

At the mean: a 10-point increase in attendance raises the standardized score by 0.078 sd.

Key insight:
The effect of attendance is **stronger** for students with higher prior GPA.

Students who were already doing well benefit more from going to class — the interaction $\mathit{priGPA} \cdot \mathit{atndrte}$ captures this heterogeneity.

Interpreting the Attendance Interaction

Testing the partial effect at the mean:

To get se of the effect at $\overline{priGPA} = 2.59$:

1. Replace $priGPA \cdot atndrte$ with $(priGPA - 2.59) \cdot atndrte$
2. Re-run the regression
3. The new coefficient on *atndrte* equals the partial effect at the mean, with its own se

Result: $\hat{\beta} = 0.0078$, $\mathrm{se} = 0.0026$, $t = 3.0$.
Statistically significant at the 1% level.

Key lessons from this example:

1. Looking at $\hat{\beta}_1 = -0.0067$ alone would suggest attendance *hurts* performance — but this is the effect when $priGPA = 0$ (meaningless!)
2. The individual t stats on $\hat{\beta}_1$ and $\hat{\beta}_6$ are both insignificant, but the F-test for joint significance rejects H_0: $\beta_1 = 0, \beta_6 = 0$ at 5%
3. **Always evaluate interactions at interesting values**

Average Partial Effects and Centering (§6-2d)

Average Partial Effects and Centering: Motivation & Setup

Motivation. Once the partial effect varies across people (quadratics, interactions), *which* number do we report? And why do the coefficients on the level terms look so strange after adding an interaction? Both puzzles are solved by averaging — and by centering variables before interacting them.

Setup. The **average partial effect (APE)** averages the observation-specific effect over the sample, $\text{APE}_j = n^{-1} \sum_i \partial \hat{y}_i / \partial x_j$. Centering each variable at its mean before forming the interaction makes the coefficient on the level term read directly as the APE. **Plan:** defining the APE, then centering.

Average Partial Effects (APE)

Average Partial Effect

In a model with quadratics and interactions, the partial effect of x_j on y varies across observations.

The **Average Partial Effect (APE)** averages the partial effect across all n observations:

$$\widehat{\text{APE}}_j = \frac{1}{n}\sum_{i=1}^{n} \frac{\partial \hat{y}_i}{\partial x_j}\bigg|_{x_i}$$

APE in the Attendance Model

Partial effect of *atndrte*: $\hat{\beta}_1 + \hat{\beta}_6\, priGPA_i$

$\widehat{\text{APE}}_{atndrte} = \hat{\beta}_1 + \hat{\beta}_6\, \overline{priGPA}$

$= -0.0067 + 0.0056(2.59) = 0.0078$

APE for *priGPA*

Partial effect: $\hat{\beta}_2 + 2\hat{\beta}_4\, priGPA_i + \hat{\beta}_6\, atndrte_i$

$\widehat{\text{APE}}_{priGPA} = \hat{\beta}_2 + 2\hat{\beta}_4\, \overline{priGPA} + \hat{\beta}_6\, \overline{atndrte}$

Centering Variables Before Creating Interactions

Consider: $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2 + u$

β_2 is the effect of x_2 when $x_1 = 0$ — often uninteresting.

Reparameterize: replace $x_1 x_2$ with $(x_1 - \bar{x}_1)(x_2 - \bar{x}_2)$:

$$y = \alpha_0 + \delta_1 x_1 + \delta_2 x_2 + \beta_3 (x_1 - \bar{x}_1)(x_2 - \bar{x}_2) + u$$

Now $\delta_2 = \beta_2 + \beta_3 \bar{x}_1$ is the APE of x_2!

Benefits of centering:

1. Coefficients on x_1 and x_2 become the APE directly
2. Standard errors for APE come automatically
3. Reduces multicollinearity between x_j and $x_j \cdot x_k$
4. The interaction coefficient β_3 is unchanged

Same logic applies to quadratics:
$y = \gamma_0 + \gamma_1 x + \beta_2 (x - \bar{x})^2 + u$
Now γ_1 is the APE of x.

Adjusted R^2 and Model Selection (§6-3)

Adjusted R^2 and Model Selection: Motivation & Setup

Motivation. R^2 *never* falls when you add a regressor — so it cannot tell you whether a variable belongs. We need a measure that **penalizes** complexity, a principled way to choose between competing models, and the judgment to know when *not* to add a control.

Setup. The **adjusted** R^2 replaces sample sums with degrees-of-freedom-corrected estimators, $\bar{R}^2 = 1 - \dfrac{\hat{\sigma}^2}{\text{SST}/(n-1)}$, so it can *fall* when a useless regressor is added. **Plan:** $\bar{R}^2$ and its properties, choosing between nonnested models, over-controlling, and adding regressors to cut the error variance.

Adjusted R-Squared: Definition

Recall: $R^2 = 1 - \text{SSR}/\text{SST}$ never falls when a variable is added.
To penalize model complexity, replace SSR/n with $\text{SSR}/(n-k-1)$:

Adjusted R-Squared (eq. 6.21)

$$\bar{R}^2 = 1 - \frac{\text{SSR}/(n-k-1)}{\text{SST}/(n-1)} = 1 - \frac{\hat{\sigma}^2}{\text{SST}/(n-1)}$$

Alternative formula: $\bar{R}^2 = 1 - (1 - R^2)\dfrac{n-1}{n-k-1}$

Why $n-k-1$?

$R^2 = 1 - \text{SSR}/n \div \text{SST}/n$. Replacing with unbiased estimators: $\hat{\sigma}^2 = \text{SSR}/(n-k-1)$ and $\text{SST}/(n-1)$ is the sample variance of y. Using unbiased estimators instead of n-scaled versions gives $\bar{R}^2$.

Adjusted R-Squared: Key Properties

Properties of $\bar{R}^2$:

1. Penalizes adding variables: $\bar{R}^2$ **can decrease** when a variable is added
2. $\bar{R}^2$ increases $\iff$ the t statistic on the new variable exceeds 1 in absolute value
3. For a group of variables: $\bar{R}^2$ increases $\iff$ the F statistic > 1
4. $\bar{R}^2 \leq R^2$ always; with small n and large k, $\bar{R}^2$ can be **negative**
5. $\bar{R}^2 < 0$ signals a very poor model relative to its complexity

Important: The F statistic for testing exclusion restrictions uses R^2, **not** $\bar{R}^2$. The formula with $\bar{R}^2$ is *not valid*!

$\bar{R}^2$ Can Be Negative

Numerical Example

If $R^2 = 0.10$, $n = 51$, $k = 10$:

$$\bar{R}^2 = 1 - (1 - 0.10)\frac{50}{40} = 1 - 1.125 = -0.125$$

A negative $\bar{R}^2$ tells us the model has too many regressors relative to the sample size and explanatory power.

Important reminders:

- The F statistic in (4.41) uses R^2, **not** $\bar{R}^2$. Using $\bar{R}^2$ in the F formula is *not* valid!
- When $\bar{R}^2$ is reported alongside R^2, remember that sometimes the $\bar{R}^2$ column is mislabeled as R^2 — always check
- $\bar{R}^2$ is most useful for comparing models with the **same dependent variable** but different regressors

Choosing Between Nonnested Models

Nested models: one is a special case of the other $\Rightarrow$ use the F test.

Nonnested models: neither is a special case of the other $\Rightarrow$ F test does not apply.

Baseball Salary (MLB1)

Model A: $\log(salary)$ on *years*, *gamesyr*, *bavg*, *hrunsyr*
Model B: same, but with *rbisyr* instead of *hrunsyr*

$\bar{R}^2_A = 0.6211$
$\bar{R}^2_B = 0.6226$

Slight preference for Model B, but the difference is tiny.

Using $\bar{R}^2$ for nonnested models:

Pick the model with the higher $\bar{R}^2$.

This works when:

1. Same dependent variable (same SST)
2. Same sample (n must be identical)

Critical limitation. You cannot use R^2 or $\bar{R}^2$ to compare y vs. $\log(y)$ as dependent variables — they measure variation in *different* things. Section 6-4 gives the fix.

Example 6.4 — CEO Compensation: Level vs. Log

Two Models for CEO Salary (CEOSAL2)

Model A (level–level):

$$\widehat{salary} = \underset{(223.90)}{830.63} + \underset{(0.0089)}{0.0163}\, sales + \underset{(11.08)}{19.63}\, roe$$

$$n = 209, \quad R^2 = 0.029, \quad \bar{R}^2 = 0.020$$

Model B (log–log):

$$\widehat{lsalary} = \underset{(0.29)}{4.36} + \underset{(0.033)}{0.275}\, lsales + \underset{(0.0040)}{0.0179}\, roe$$

$$n = 209, \quad R^2 = 0.282, \quad \bar{R}^2 = 0.275$$

Cannot compare $R^2 = 0.029$ (for *salary*) with $R^2 = 0.282$ (for $\log(salary)$) — they explain variation in **different variables**. We need the method from Section 6-4 to make a valid comparison.

Over-Controlling: Adding Bad Controls

It is possible to control for *too many* variables: including a "bad control" can introduce bias or mask the very effect you want to measure.

Beer Taxes and Traffic Fatalities

Want: effect of beer *tax* on *fatalities*.

Good controls: *miles*, *percmale*, $perc16\_21$ (demographics, driving exposure)

Bad control: *beercons* (beer consumption)
If we hold *beercons* fixed, how would *tax* affect fatalities? Only through non-alcohol channels — not what we want!

When is a control "bad"?

A variable is a bad control if it is:

1. On the **causal pathway** between x and y (a "mediator")
2. An **outcome** of the treatment variable
3. Would change the **ceteris paribus interpretation** to something uninteresting

Rule of thumb: Think carefully about the ceteris paribus interpretation. Does holding this variable fixed make economic sense?

Adding Regressors to Reduce Error Variance

The tradeoff: Adding a variable to a regression:

+ Reduces error variance σ^2 (takes variation out of *u*)
– May increase multicollinearity (inflates $\mathrm{Var}(\hat{\beta}_j)$)

Clear case: If new variables are **uncorrelated** with the regressors of interest:

- No multicollinearity increase
- Error variance decreases
- Standard errors of all coefficients shrink
- **Always include** such variables

Computer Equipment Grants

Effect of randomly assigned grant amount on college GPA.

- Random assignment $\Rightarrow$ grant is uncorrelated with HS GPA, family background, etc.
- Including HS GPA and test scores as controls: reduces $\hat{\sigma}^2$ without increasing multicollinearity
- Result: more precise estimate of the grant effect

With random assignment: include pre-treatment controls to improve precision, even though they are not needed for unbiasedness.

Prediction and Residual Analysis (§6-4)

Prediction and Residual Analysis: Motivation & Setup

Motivation. So far we used regression to estimate ceteris paribus *effects*. But we also want to **predict** y at chosen values of the regressors — and to be honest about the uncertainty, which differs for an *average* outcome versus a *single* new unit.

Setup. Distinguish the **confidence interval** for the mean $\mathbb{E}(y \mid \mathbf{x}_0)$ from the wider **prediction interval** for an individual y_0 (which adds the error variance $\hat{\sigma}^2$). **Plan:** confidence intervals for predictions, prediction intervals, residual analysis, and recovering y when $\log(y)$ is the dependent variable.

Confidence Intervals for Predictions

We often want the predicted *mean* response at chosen values $x_j = c_j$, with a confidence interval around it. The point estimate is immediate; its standard error takes a small trick.

Given $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \cdots + \hat{\beta}_k x_k$, we predict $\mathbb{E}[y \mid x_1 = c_1, \ldots, x_k = c_k]$:

$$\hat{\theta}_0 = \hat{\beta}_0 + \hat{\beta}_1 c_1 + \cdots + \hat{\beta}_k c_k$$

How to get $\mathrm{se}(\hat{\theta}_0)$?

The reparameterization trick:

1. Define $x_j^0 = x_j - c_j$ for each variable
2. Regress y on $x_1^0, \ldots, x_k^0$
3. The new intercept $= \hat{\theta}_0$ with its own se!

95% CI: $\hat{\theta}_0 \pm t_{0.025} \cdot \mathrm{se}(\hat{\theta}_0)$

Why this works:

Writing $\beta_0 = \theta_0 - \beta_1 c_1 - \cdots - \beta_k c_k$ and substituting:

$y = \theta_0 + \beta_1(x_1 - c_1) + \cdots + \beta_k(x_k - c_k) + u$

The intercept in this new regression is exactly θ_0, the expected value of y at the specified x-values.

Note: The variance of the prediction is smallest when $c_j = \bar{x}_j$ for all j (predicting at the sample means). Predictions far from the means have wider CIs.

Example 6.5 — Confidence Interval for Predicted College GPA

College GPA Prediction (GPA2 — eq. 6.32)

$$\widehat{colgpa} = \underset{(0.075)}{1.493} + \underset{(0.00007)}{0.00149}\, sat - \underset{(0.00056)}{0.01386}\, hsperc$$

$$-\underset{(0.01650)}{0.06088}\, hsize + \underset{(0.00227)}{0.00546}\, hsize^2$$

$$n = 4{,}137, \quad R^2 = 0.278, \quad \bar{R}^2 = 0.277, \quad \hat{\sigma} = 0.560$$

Predict for: $sat = 1{,}200$, $hsperc = 30$, $hsize = 5$

$\widehat{colgpa} \approx 2.70$

Using the reparameterization trick:
$\mathrm{se}(\hat{\theta}_0) = 0.020$

95% CI for expected GPA:

$$2.70 \pm 1.96(0.020) = [2.66, 2.74]$$

Very narrow! With $n = 4{,}137$, the expected value is precisely estimated.

This is a CI for the **average** GPA of all students with these characteristics.

Prediction Intervals vs. Confidence Intervals

Two Types of Intervals

Confidence interval: for $\mathbb{E}[y \mid \mathbf{x}]$ (the average outcome for a subpopulation)

Prediction interval: for y^0 (a specific individual's future outcome)

The prediction error is $\hat{e}^0 = y^0 - \hat{y}^0$. The new unit's error u^0 is independent of the estimation sample, so $\mathrm{Cov}(\hat{y}^0, u^0) = 0$ and the two variances simply add:

$$\mathrm{Var}(\hat{e}^0) = \mathrm{Var}(\hat{y}^0) + \sigma^2.$$

Two sources of uncertainty:

1. Sampling error in $\hat{\beta}_j$'s ($\to 0$ as $n \to \infty$)
2. Error variance σ^2 (does **not** shrink with n)

Standard error of prediction:

$$\mathrm{se}(\hat{e}^0) = \sqrt{[\mathrm{se}(\hat{y}^0)]^2 + \hat{\sigma}^2}$$

95% prediction interval:

$$\hat{y}^0 \pm t_{0.025} \cdot \mathrm{se}(\hat{e}^0)$$

The prediction interval is **always wider** than the confidence interval because it includes $\hat{\sigma}^2$.

For large n: $\mathrm{se}(\hat{y}^0)$ is tiny, so $\mathrm{se}(\hat{e}^0) \approx \hat{\sigma}$ — the width is dominated by the irreducible error variance.

Example 6.6 — Prediction Interval for Future College GPA

Individual Prediction (GPA2)

Same student: *sat* $= 1{,}200$, *hsperc* $= 30$, *hsize* $= 5$.

$\hat{y}^0 = 2.70$, $\mathrm{se}(\hat{y}^0) = 0.020$, $\hat{\sigma} = 0.560$

$$\mathrm{se}(\hat{e}^0) = \sqrt{(0.020)^2 + (0.560)^2} \approx 0.560$$

95% prediction interval: $2.70 \pm 1.96(0.560) = [1.60, 3.80]$

Compare:

- CI for average GPA: $[2.66, 2.74]$ (very narrow)
- PI for individual GPA: $[1.60, 3.80]$ (very wide!)

The prediction interval spans more than 2 full GPA points.

Why so wide?

The model explains only 27.8% of the variation in college GPA ($R^2 = 0.278$).

Unobserved factors (motivation, study habits, health, social environment) account for most of the variation.

Good news: SAT and HS rank do not predetermine college success!

Residual Analysis

The residual itself carries information: a large $|\hat{u}_i|$ flags an observation the model badly misses — an underpriced house, a value-adding school, or an over- or under-paid athlete.

Residual Analysis

Examine individual residuals $\hat{u}_i = y_i - \hat{y}_i$ to identify observations that are unusually above or below the regression line.

Applications:

- **Housing:** The house with the most negative residual is the most "underpriced" relative to its predicted value
- **Law schools:** Rank schools by residual after controlling for inputs (LSAT, GPA) — the highest residual school adds the most "value"
- **Sports:** Identify overpaid/underpaid athletes relative to their performance

Housing Prices (HPRICE1)

Regress *price* on *lotsize*, *sqrft*, *bdrms*.
$n = 88$ homes.

The most negative residual: $\hat{u}_{81} = -120{,}206$.
House 81's asking price is $120,206 below its predicted price.

Possible explanations:

- Unobserved defects (bad neighborhood, needs repairs)
- Motivated seller
- Genuinely underpriced (a "bargain")

The residual reflects everything **not** captured by the model. Large residuals may reveal model misspecification or unusual observations.

Predicting y When $\log(y)$ Is the Dependent Variable

If the model is $\log(y) = \beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k + u$, OLS predicts $\widehat{\log(y)}$, not y directly.

Naive prediction: $\tilde{y} = \exp(\widehat{\log(y)})$

> This **systematically underestimates** $\mathbb{E}[y \mid \mathbf{x}]$ because $\mathbb{E}[\exp(u)] > \exp(\mathbb{E}[u]) = 1$ by Jensen's inequality whenever $\mathrm{Var}(u) > 0$.

Correction Under Normality (eq. 6.40)

If $u \mid \mathbf{x} \sim \mathcal{N}(0, \sigma^2)$, then $\mathbb{E}[\exp(u)] = \exp(\sigma^2/2)$, so:

$$\hat{y} = \exp\left(\frac{\hat{\sigma}^2}{2}\right) \cdot \exp\left(\widehat{\log(y)}\right)$$

where $\hat{\sigma}^2 = \text{SSR}/(n - k - 1)$ is the squared SER. The correction factor $\exp(\hat{\sigma}^2/2) > 1$ inflates the naive prediction.

Predicting y From $\log(y)$: Smearing Estimate

Smearing Estimate (eq. 6.43)

Without assuming normality, estimate $\mathbb{E}[\exp(u)]$ from residuals:

$$\hat{\alpha}_0 = \frac{1}{n}\sum_{i=1}^{n} \exp(\hat{u}_i)$$

Then the adjusted prediction is:

$$\hat{y} = \hat{\alpha}_0 \cdot \exp\left(\widehat{\log(y)}\right)$$

Since $\exp$ is convex, $\hat{\alpha}_0 > 1$ always.

Which correction to use?

- Errors approximately normal $\Rightarrow$ use eq. 6.40 ($\exp(\hat{\sigma}^2/2)$); more efficient when correct
- Unsure about normality $\Rightarrow$ use smearing estimate (eq. 6.43); robust to non-normality
- Both are consistent estimators of $\mathbb{E}[y \mid \mathbf{x}]$

Example 6.7 — Predicting CEO Salaries

CEO Salary Prediction (CEOSAL2 — eq. 6.45)

$$\widehat{lsalary} = \underset{(0.257)}{4.504} + \underset{(0.039)}{0.163}\, lsales + \underset{(0.050)}{0.109}\, lmktval + \underset{(0.0053)}{0.0117}\, ceoten$$

$$n = 177, \quad R^2 = 0.318$$

Predict for: $sales = 5{,}000$ (millions),
$mktval = 10{,}000$ (millions), $ceoten = 10$ years.

$$\widehat{lsalary} = 4.504 + 0.163\log(5{,}000) + 0.109\log(10{,}000) + 0.0117(10) \approx 7.013$$

Naive: $\exp(7.013) = \$1{,}110{,}983$

Adjusted predictions:

- Smearing: $\hat{\alpha}_0 = 1.136$
  $\hat{y} = 1.136 \times 1{,}110{,}983 = \$1{,}262{,}077$
- Normality: $\exp(\hat{\sigma}^2/2) = 1.117$
  $\hat{y} = 1.117 \times 1{,}110{,}983 = \$1{,}240{,}968$

Both adjustments add $\approx$ \$130,000–\$150,000 to the naive prediction. The adjustment matters!

Comparing Models with Different Dependent Variables

Problem: Cannot compare R^2 from a model for y with R^2 from a model for $\log(y)$.

Solution: Compute a "pseudo-R^2" for y from the log model:

$$\tilde{R}^2 = 1 - \frac{\sum_{i=1}^{n} \hat{r}_i^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}$$

where $\hat{r}_i = y_i - \hat{\alpha}_0 \exp(\widehat{\log(y)}_i)$ are residuals for predicting y (not $\log y$).

Example 6.8: CEO Salaries

From the log model: $\mathrm{Corr}(salary_i, \hat{m}_i)^2 = (0.493)^2 = 0.243$

From the level model (eq. 6.49): $R^2 = 0.201$

The log model explains **more** variation in *salary* (24.3% vs. 20.1%) and has more interpretable coefficients.

Conclusion. The log model wins on both goodness-of-fit and interpretability for CEO salary.

Chapter 6 Summary: Scaling and Functional Form

Data Scaling (§6-1)

- Rescaling changes $\hat{\beta}$ and SEs proportionally
- t, F, R^2 are **invariant** to rescaling
- Beta coefficients: standardize all variables for cross-regressor comparison

Functional Form (§6-2)

- Four log combinations; exact % change $= \exp(\cdot) - 1$
- Quadratics: marginal effect $= \hat{\beta}_1 + 2\hat{\beta}_2 x$
- Interactions: partial effect depends on other variable
- APE: average partial effect over sample
- Centering at $\bar{x}$: constant $=$ APE directly

Chapter 6 Summary: Model Selection and Prediction

Adjusted R^2 (§6-3)

- Penalizes extra regressors; can be negative
- Valid for nonnested models with **same** y
- Cannot compare models with different y's
- Beware over-controlling (bad controls)

Prediction (§6-4)

- CI for $\mathbb{E}[y|\mathbf{x}]$: narrow (sampling uncertainty)
- PI for new y^0: wide (adds $\hat{\sigma}^2$ for error)
- Log models: correct with $\exp(\hat{\sigma}^2/2)$ or smearing estimate
- Compare y vs. $\log(y)$ using pseudo-R^2

Core theme of Ch. 6: OLS is more flexible than it looks. Scaling, functional form, and prediction all build on the same estimator — what changes is *how we set up the model* and *how we read the results*.

The Bootstrap: Motivation

Problem: Formulas for standard errors are sometimes

- Hard to derive analytically
- Not good approximations to the true sampling variation
- Unavailable for complex estimators

Solution: Use a **resampling method** — treat the observed data as a population from which we can draw new samples.

> The most common resampling method is the **nonparametric bootstrap**. It requires no distributional assumptions.

The Bootstrap Idea

1. We have an estimator $\hat{\theta}$ computed from a sample of size n
2. Treat the n observations as the "population"
3. Draw n observations **with replacement** to create a bootstrap sample
4. Recompute $\hat{\theta}^{(b)}$ on each bootstrap sample
5. Repeat m times and use the variation in $\{\hat{\theta}^{(b)}\}$ to measure uncertainty

Bootstrap Standard Error: Algorithm and Formula

Algorithm:

1. List observations $1, \ldots, n$
2. Draw n integers randomly with replacement $\Rightarrow$ bootstrap sample b (size n)
3. Compute estimate $\hat{\theta}^{(b)}$ on sample b
4. Repeat steps 2–3 a total of m times
5. Compute $\bar{\theta} = m^{-1} \sum_{b=1}^{m} \hat{\theta}^{(b)}$
6. Compute $\text{bse}(\hat{\theta})$ from eq. [6.50]

Choice of m: Typical value is $m = 1{,}000$.
Even $m = 500$ often gives a reliable SE.
m has nothing to do with the sample size n.

Bootstrap Standard Error (eq. 6.50)

$$\text{bse}(\hat{\theta}) = \left[(m-1)^{-1} \sum_{b=1}^{m} \left(\hat{\theta}^{(b)} - \bar{\theta} \right)^2 \right]^{1/2}$$

where $\bar{\theta} = m^{-1} \sum_{b=1}^{m} \hat{\theta}^{(b)}$ is the average of all bootstrap estimates.

$\text{bse}(\hat{\theta})$ is the **sample standard deviation** of the m bootstrap estimates. It is a consistent estimator of $\text{se}(\hat{\theta})$.

Bootstrap in Practice

When to use the bootstrap:

- Asymptotic SE formula unavailable (e.g. nonlinear functions of $\hat{\beta}$)
- Small samples where asymptotics may be poor
- Checking whether analytical SEs are reliable

Going further:
Bootstrap samples can be used to compute *p*-values for *t* and *F* statistics, or to construct confidence intervals directly — often more accurate than bse$(\hat{\theta})$ in a *t* statistic.

See Horowitz (2001).

R: Bootstrap SE for OLS coefficient

```
library(wooldridge); library(boot)
data(wage1)
boot_fn <- function(data, idx) {
  m <- lm(log(wage) ~ educ + exper
    + I(exper^2) + tenure,
    data = data[idx, ])
  coef(m)["educ"] }
set.seed(42)
b <- boot(wage1, boot_fn, R=1000)
sd(b$t)   # bootstrap SE
```

R Code: Scaling and Beta Coefficients

```r
library(wooldridge)

## --- Scaling (Table 6.1) ---
data(bwght)
m1 <- lm(bwght ~ cigs + faminc, data = bwght)
bwght$bwghtlbs <- bwght$bwght / 16
m2 <- lm(bwghtlbs ~ cigs + faminc, data = bwght)
# Compare: coef(m2) = coef(m1)/16; t-stats identical

## --- Beta Coefficients (Example 6.1) ---
data(hprice2)
m_std <- lm(scale(price) ~ scale(nox) + scale(crime)
  + scale(rooms) + scale(dist) + scale(stratio),
  data = hprice2)
coef(m_std)  # beta coefficients (intercept ~ 0)
```

R Code: Quadratic Models

```
## --- Quadratic (WAGE1 eq. 6.12) ---
data(wage1)
m_quad <- lm(wage ~ exper + I(exper^2), data = wage1)
summary(m_quad)
tp <- abs(coef(m_quad)["exper"]) /
  (2 * abs(coef(m_quad)["I(exper^2)"]))
cat("Turning point:", round(tp, 1), "years\n")

## --- Interaction (Example 6.3: ATTEND) ---
data(attend)
m_int <- lm(stndfnl ~ atndrte + priGPA + ACT +
  I(priGPA^2) + I(ACT^2) + atndrte:priGPA,
  data = attend)
summary(m_int)
# APE of atndrte at mean priGPA
ape <- coef(m_int)["atndrte"] +
  coef(m_int)["atndrte:priGPA"] * mean(attend$priGPA)
```

R Code: Predictions and Log Models

```
## --- Prediction CI (Example 6.5: GPA2) ---
data(gpa2)
m_gpa <- lm(colgpa ~ sat + hsperc + hsize + I(hsize^2),
  data = gpa2)
nd <- data.frame(sat=1200, hsperc=30, hsize=5)
predict(m_gpa, newdata=nd, interval="confidence")
predict(m_gpa, newdata=nd, interval="prediction")

## --- Log model prediction (Example 6.7: CEOSAL2) ---
data(ceosal2)
m_log <- lm(lsalary ~ lsales + lmktval + ceoten,
  data = ceosal2)
sigma_hat <- summary(m_log)$sigma
alpha_norm  <- exp(sigma_hat^2 / 2)       # normality
alpha_smear <- mean(exp(residuals(m_log))) # smearing
```